

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – VII

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Class: SYBSc IT	Subject: DATA STRUCTURE
Date of Assignment: 05/08/2026	Date/Time of Submission: 05/08/2026

Hash Table:

Write a program to:

a. Implement a hash table with separate chaining for collision handling.

CODE:

```
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)] .append((key,value))

insert(10,"STRAWBERRY")
insert(30,"LICHI")
insert(20,"PAPAYA")
insert(15,"BLUEBERRY")
insert(48,"APPLE")

print(table)
print("IQRA S021")
```

OUTPUT:

```
===== RESTART: D:/DATA STRUCTURE SYIT/PRACTICAL7.py =====
[[ (10, 'STRAWBERRY'), (30, 'LICHI'), (20, 'PAPAYA'), (15, 'BLUEBERRY') ], [], [], [(48, 'APPLE')], []]
IQRA S021
```

b. Store and retrieve data from the hash table.

CODE:

```

size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)] .append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not found"

insert(10,"STRAWBERRY")
insert(30,"LICHI")
insert(20,"PAPAYA")
insert(15,"BLUEBERRY")
insert(48,"APPLE")

print(table)

key = int(input("Enter your key value: "))
print(search(key))
|
print("IQRA S021")

```

OUTPUT:

```

===== RESTART: D:/DATA STRUCTURE SYIT/PR
[[ (10, 'STRAWBERRY'), (30, 'LICHI'), (20, 'PAPAYA'), (15, 'BLUEBERRY') ], [], [], [(48, 'APPLE')], []]
Enter your key value: 48
APPLE
IQRA S021
|

===== RESTART: D:/DATA STRUCTURE SYIT/PR
[[ (10, 'STRAWBERRY'), (30, 'LICHI'), (20, 'PAPAYA'), (15, 'BLUEBERRY') ], [], [], [(48, 'APPLE')], []]
Enter your key value: 10
STRAWBERRY
IQRA S021

===== RESTART: D:/DATA STRUCTURE SYIT/PR
[[ (10, 'STRAWBERRY'), (30, 'LICHI'), (20, 'PAPAYA'), (15, 'BLUEBERRY') ], [], [], [(48, 'APPLE')], []]
Enter your key value: 88
Not found
IQRA S021

```